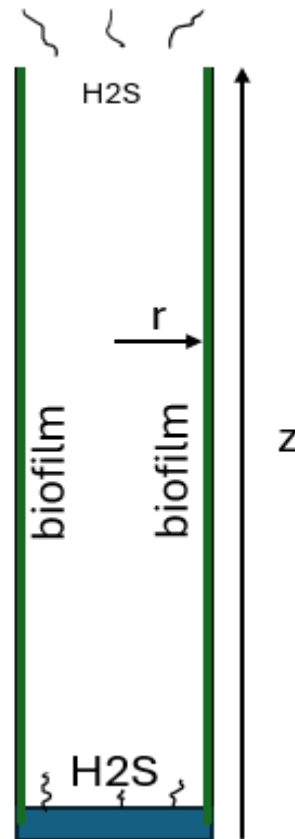
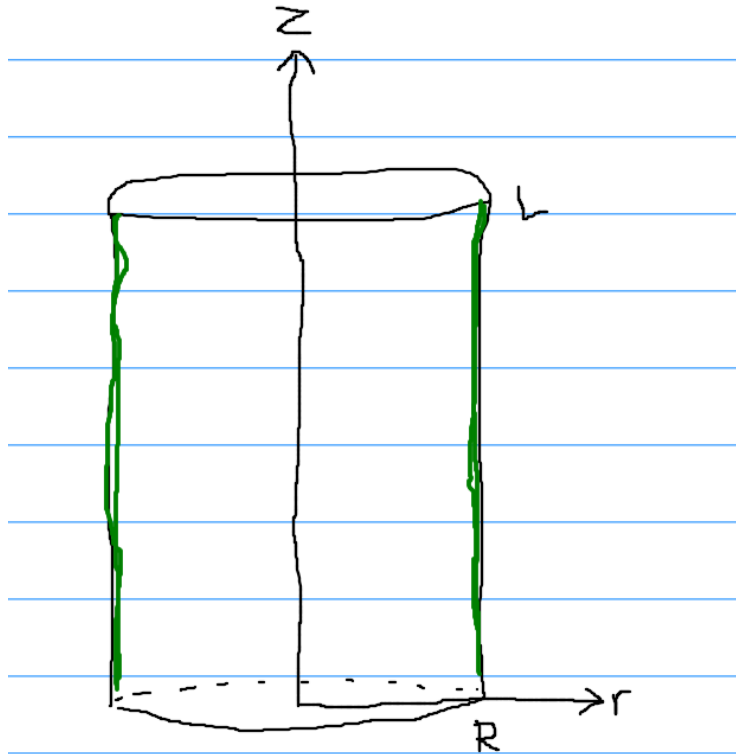


Plan

1. Short intro to 2D models (5 min)
2. Exercise on 2D model with heat bed (1 hour)
3. Break (5-10 min)
4. Example of 2D model in COMSOL (10-20 minutes)

2D model example



Cylindrical 2D

$$\frac{\partial C_A}{\partial t} = D \left(\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial C_A}{\partial r} \right) + \frac{\partial^2 C_A}{\partial z^2} \right) + R_A$$

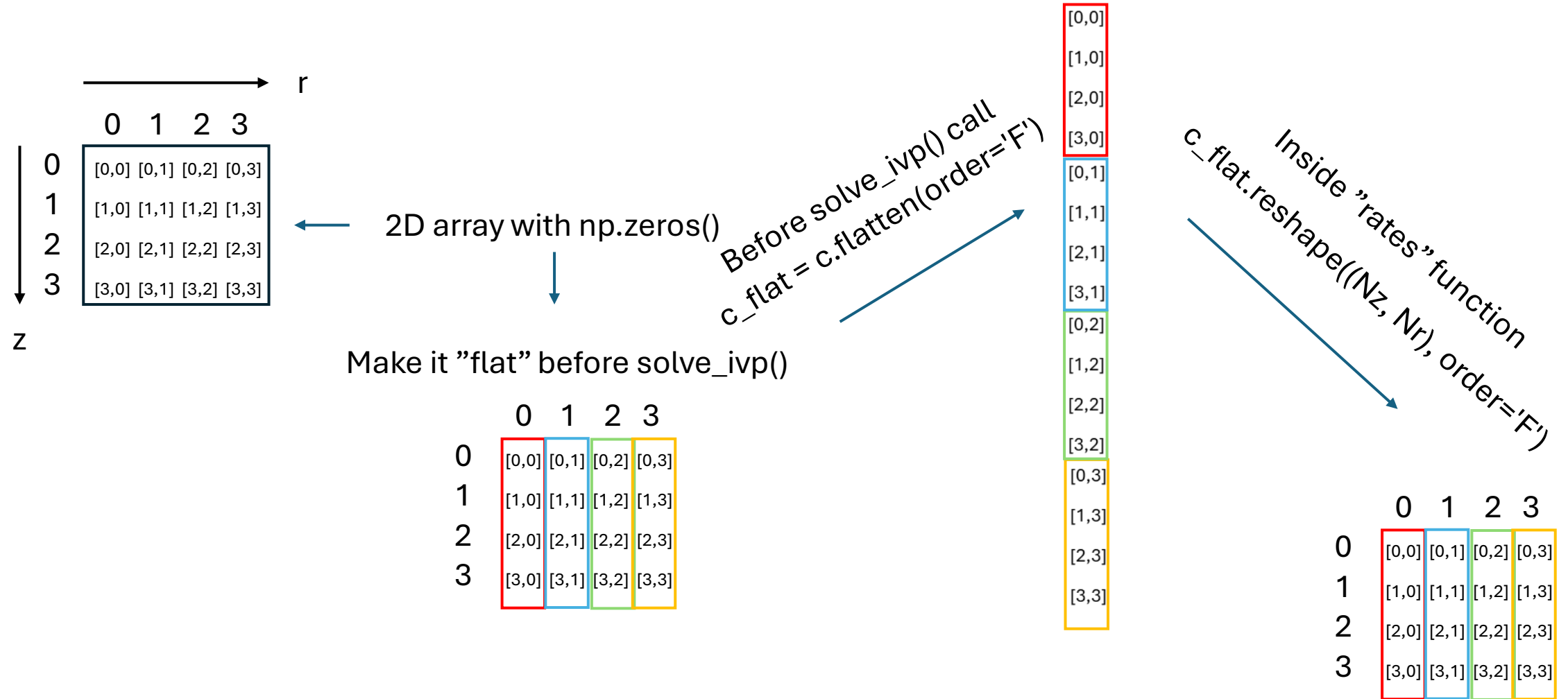
$\downarrow$ 1 IC $\downarrow$ 2 BC in r dim $\downarrow$ 2 BC in z dim

$$-R_A = k \cdot C_A \text{ (for simplicity)}$$

By product rule of differentiation

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial C_A}{\partial r} \right) = \frac{\partial^2 C_A}{\partial r^2} + \frac{1}{r} \frac{\partial C_A}{\partial r}$$

`solve_ivp()` expects a 1D array, but we intuitively understand the problem as 2D



Governing equation for diffusion only in 2D rectangular coordinates

Interior points: $\frac{\partial C[i,j]}{\partial t} = D \left(\frac{c[i+1,j] - 2c[i,j] + c[i-1,j]}{\Delta z^2} + \frac{c[i,j+1] - 2c[i,j] + c[i,j-1]}{\Delta r^2} \right)$

Ghost point top



At top:

$$\frac{\partial C[Ni,j]}{\partial t} = D \left(\frac{c[Ni+1,j] - 2c[Ni,j] + c[Ni-1,j]}{\Delta z^2} + \frac{c[Ni,j+1] - 2c[Ni,j] + c[Ni,j-1]}{\Delta r^2} \right)$$

At right:

$$\frac{\partial C[i,Nj]}{\partial t} = D \left(\frac{c[i+1,Nj] - 2c[i,Nj] + c[i-1,Nj]}{\Delta z^2} + \frac{c[i,Nj+1] - 2c[i,Nj] + c[i,Nj-1]}{\Delta r^2} \right)$$



Ghost point right

top right corner:

$$\frac{\partial C[Ni,Nj]}{\partial t} = D \left(\frac{c[Ni+1,Nj] - 2c[Ni,Nj] + c[Ni-1,Nj]}{\Delta z^2} + \frac{c[Ni,Nj+1] - 2c[Ni,Nj] + c[Ni,Nj-1]}{\Delta r^2} \right)$$



Ghost point top



Ghost point right

